Probability

Year 4 · **Lesson Worksheet**

Worked Examples

Probability as fraction/decimal/percentage: A bag contains 4 red, 3 blue and 5 green counters. A counter is picked at random. Write the probability it is blue as a fraction, decimal and percentage.

Answer: There are 12 counters in total, 3 of which are blue, so P(blue) = 3/12 = 1/4 = 0.25 = 25%.

Probability of NOT happening: The probability that it rains tomorrow is 0.35. What is the probability that it does NOT rain?

Answer: P(not rain) = 1 − 0.35 = 0.65.

Combined events: A fair coin is flipped twice. What is the probability of getting two heads?

Answer: List outcomes: HH, HT, TH, TT (4 equally likely outcomes). Only 1 is HH, so P(two heads) = 1/4.

Probability Basics (15 questions)

1. A fair six-sided dice is rolled once. What is the probability of rolling a 4? Give your answer as a fraction.
2. A spinner has 8 equal sections numbered 1 to 8. What is the probability of landing on an odd number?
3. Write the probability 'certain' as a number. A) 0 B) 0.25 C) 0.5 D) 0.75 E) 1
4. A bag contains 10 balls, 6 of which are yellow. What is the probability of picking a yellow ball, as a decimal?
5. Order these words from least likely to most likely: likely, impossible, evens, certain, unlikely.
6. A coin is flipped once. What is the probability of landing on tails, as a percentage?
7. Which word best describes an event with probability 0.5? A) Impossible B) Unlikely C) Evens D) Likely E) Certain
8. A bag contains only blue counters. A counter is picked at random. What is the probability it is blue?
9. List all the possible outcomes when a single coin is flipped.
10. A box contains 3 red pens and 7 blue pens. What is the probability of picking a red pen, as a fraction in its simplest form?
11. Select ALL that apply: which of these are valid probabilities for an event? A) 0 B) 1.2 C) 0.5 D) −0.3 E) 1
12. A fair six-sided dice is rolled once. What is the probability of rolling a number greater than 4? A) 1/6 B) 1/3 C) 1/2 D) 2/3 E) 5/6
13. The probability that a train is late is 3/20. Write this probability as a percentage.
14. List all the possible outcomes when a single fair six-sided dice is rolled.
15. A spinner is divided into 4 equal sections coloured red, blue, green and yellow. What is the probability of NOT landing on red?

Applying Probability (15 questions)

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1. A bag contains 5 green, 3 red and 2 yellow counters. A counter is picked at random. What is the probability it is NOT green? Give your answer as a fraction.

2. A spinner has 10 equal sections: 4 are blue, 6 are white. What is the probability of landing on white, as a decimal?

3. The probability that Amir wins a game is 0.4. What is the probability that he loses (assuming no draws)? A) 0.2 B) 0.4 C) 0.5 D) 0.6 E) 0.8

4. A fair coin is flipped twice. List all 4 possible outcomes, then state the probability of getting exactly one head and one tail.

5. A bag contains 20 sweets: 8 are orange, the rest are lemon. What is the probability of picking a lemon sweet, as a percentage?

6. Two fair coins are flipped. What is the probability of getting at least one head? A) 1/4 B) 1/2 C) 3/4 D) 1 E) 0

7. A bag has 4 red and 6 blue counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is red, what is the probability the second counter picked is also red?

8. Select ALL that apply: a fair six-sided dice is rolled once. Which statements are true? A) P(rolling a 7) = 0. B) P(rolling an even number) = 1/2. C) P(rolling a number less than 7) = 1. D) P(rolling a 1) = 1/6. E) P(rolling a multiple of 5) = 1/2.

9. A box of 15 chocolates has 9 milk and 6 dark chocolates. What is the probability of picking a milk chocolate, as a fraction in its simplest form?

10. A weather forecaster says there is a 70% chance of sunshine tomorrow. What is the probability, as a fraction, that it will NOT be sunny?

11. A bag contains only red and blue counters. P(red) = 0.3. If there are 40 counters in total, how many are blue?

12. A fair spinner has sections numbered 1 to 5. It is spun twice. What is the probability that it lands on 5 both times? A) 1/25 B) 1/10 C) 2/5 D) 1/5 E) 2/25

13. A drawer contains 6 black socks and 4 white socks. One sock is pulled out at random, not put back, and a second sock is pulled out. What is the probability that both socks are black?

14. In a raffle, 200 tickets are sold and Priya buys 5 of them. What is the probability that Priya's ticket is drawn as the winner, as a decimal?

15. Select ALL that apply: a bag contains 3 red, 4 blue and 3 green counters (10 total). Which statements are true? A) P(red) = 3/10. B) P(not blue) = 3/5. C) P(green) = P(blue). D) P(blue) = 0.4. E) P(red or green) = 0.6.

Probability Challenge (15 questions)

1. In a jar of red, blue and green buttons, exactly half are red and a quarter are blue. Given that the jar holds 20 buttons altogether, how many buttons are green?

2. Roll two fair dice and add the two scores together. What is the probability that this sum equals 8? Write your answer as a fraction. A) 1/12 B) 5/36 C) 1/6 D) 1/4

3. Inside a box sit 4 red counters and 2 blue ones. One counter is drawn out and kept to one side, then a second is drawn. Find the probability that both counters drawn are blue.

4. Spin a 1-to-4 spinner once and toss a fair coin once. What is the chance that the spinner lands on an even number and the coin lands on heads, both happening together?

5. Select ALL that apply: a box contains 10 cards numbered 1 to 10, and one card is drawn at random. Which statements are true? A) P(drawing an even number) = 1/2. B) P(drawing a multiple of 5) = 1/5. C) P(drawing a number greater than 8) = 1/5. D) P(drawing a square number) = 1/5. E) P(drawing a prime number) = 2/5.

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6. A bag has 6 yellow and 3 purple counters. Two counters are picked one after another without replacement. What is the probability that the first is yellow and the second is purple? A) 1/6 B) 1/5 C) 1/4 D) 1/3 E) 2/5

7. On any day, there's a 1/4 chance the morning bus runs late, with each day's chance independent of the last. What is the probability that the bus is late on two days running?

8. A fair six-sided dice is rolled twice. What is the probability that both rolls show the same number?

9. A box holding only red and green counters has them in the ratio 1:3. One red counter is taken out and not put back. Starting from 16 counters in total, what is the probability that the next one drawn is green?

10. Two fair coins are tossed at the same time. Select ALL of these statements that are true: A) There are 4 equally likely outcomes. B) P(two heads) = 1/4. C) P(at least one head) = 3/4. D) P(exactly one tail) = 1/2. E) P(no tails) = 1/2.

11. Imagine a spinner split into four equal parts labelled 1, 2, 3 and 4. Spin it twice and add the two results. What is the probability that this sum comes out even?

12. A pouch holds 3 red counters and 2 blue counters. Someone pulls one out, keeps it aside, then pulls out another. What is the probability the two counters are not the same colour as each other? A) 2/5 B) 3/5 C) 1/2 D) 3/10

13. Half the pupils in a class of 20 own a pet. Among the pet owners, exactly half have a dog. If you pick a pupil from the whole class at random, what is the probability they own a dog?

14. On a game show, a contestant faces 4 identical boxes, only one of which hides a prize. If they choose a box without any clues, what is the probability their choice is NOT the prize box?

15. Priya spins two separate 1-2-3 spinners and adds the results together. What is the probability that her total is at least 5?

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